

Cisco Packet Tracer

Comprehensive Device & Component Reference Guide

"A structured technical reference covering all device categories, specifications, interconnections, command-line operations, and use-case scenarios within Cisco Packet Tracer's simulation environment."

Scope of Document

Networking Devices · End Devices · IoT
Components
Connection Media · WAN Emulation ·
Multiuser Systems
Legacy Equipment · CLI & Terminal
Command Reference

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Executive Summary

Cisco Packet Tracer is a network simulation platform developed by Cisco Systems, widely used in CCNA, CCNP, and general networking education. It provides a virtual environment to design, configure, and troubleshoot networks comprising routers, switches, wireless equipment, end-user devices, and Internet-of-Things (IoT) components.

This document serves as a **professional technical reference** for every device category available in Packet Tracer's device selection panel, and additionally provides a **command-line reference** spanning both Cisco IOS and standard operating system terminals. It is structured to answer four core questions:

1. **What is it?** — Functional definition and OSI layer classification.
2. **What does it do?** — Operational role within a network topology.
3. **When should it be used?** — Practical scenario-based guidance and differentiation from similar devices.
4. **How is it configured/verified?** — Relevant CLI or terminal commands.

The guide progresses from Layer 1 physical media through Layer 3 routing infrastructure, into end-user and IoT devices, before covering WAN emulation, multiuser lab environments, legacy equipment, and a dedicated command-line reference chapter. A consolidated decision matrix and appendix are provided for quick operational reference.

Document Structure Overview

- **Chapter 1:** Foundational Concepts & OSI Mapping
- **Chapter 2:** Network Infrastructure Devices (Routers, Switches, Wireless)
- **Chapter 3:** End Devices
- **Chapter 4:** IoT Components
- **Chapter 5:** Connections & Cabling Standards
- **Chapter 6:** WAN Emulation & Cloud Devices
- **Chapter 7:** Multiuser Connectivity
- **Chapter 8:** Legacy Devices
- **Chapter 9:** Networking Commands Reference — Basic to Advanced (CLI & Terminal)
- **Chapter 10:** Consolidated Decision Matrix
- **Appendices:** CLI Quick Cards, Glossary

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CHAPTER 1

Foundational Concepts and OSI Layer Mapping

1.1 Purpose of Device Classification

Packet Tracer organizes its device palette according to **functional role and OSI layer of operation**. Understanding this classification is essential before configuring any topology, as it determines:

- Which protocols a device can process (e.g., a Layer 2 switch cannot route between subnets).
- Which physical/logical interfaces are available.
- How the device should be cabled to peers.

1.2 OSI Layer Reference Table

Table 1.1: Device Category Mapping to OSI Model Layers

OSI Layer	Layer Name	Representative Devices
7–5	Application/Session	Servers, PCs, IoT Applications
4	Transport	End Devices (TCP/UDP session handling)
3	Network	Routers, Multilayer Switches, Wireless Routers
2	Data Link	Switches, Bridges, Access Points
1	Physical	Hubs, Repeaters, Cables, Fiber, Serial links

NOTE

A device may span multiple layers. For example, a Wireless Router operates at Layer 3 (routing/NAT) while also performing Layer 2 (switching) and Layer 1 (RF signal transmission) functions.

1.3 Network Topology Reference Diagram

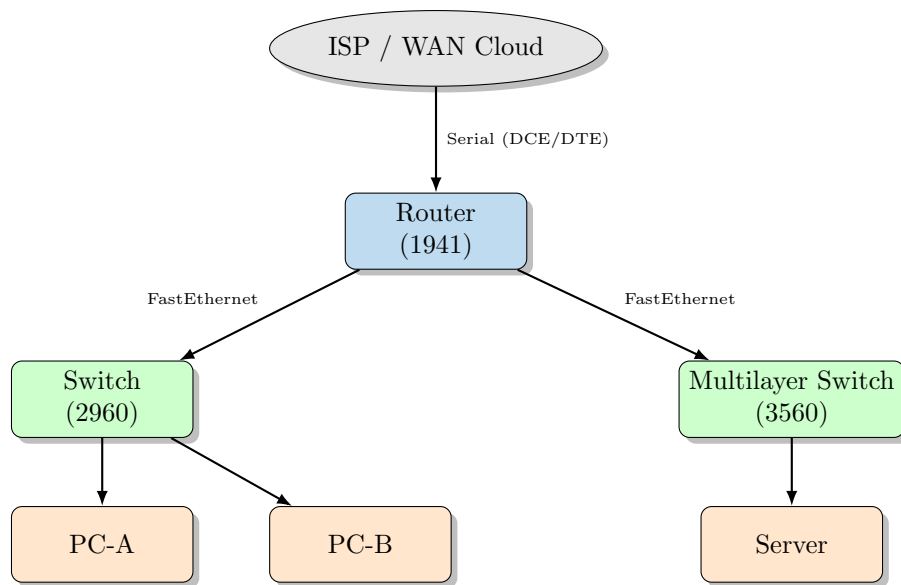


Figure 1.1: Reference topology illustrating typical device placement (WAN → Router → Switches → End Devices)

CHAPTER 2

Network Infrastructure Devices

2.1 Routers

2.1.1 Functional Overview

Routers operate at **Layer 3** of the OSI model, forwarding packets between distinct IP networks based on routing tables built through static configuration or dynamic routing protocols (RIP, OSPF, EIGRP, BGP). Every router in Packet Tracer supports:

- CLI configuration via Console, Telnet, or SSH.
- Interface types: FastEthernet, GigabitEthernet, Serial, Loopback (virtual).
- NAT, ACLs, DHCP server functionality, routing protocol configuration.

2.1.2 Router Model Specifications

Model	Series	Fixed Interfaces	Typical Use Case
1841	ISR G1	2× FastEthernet	Basic branch routing labs; static/default routing
2620XM	Classic	1× FastEthernet, modular WIC	Legacy WAN simulation with serial WICs
2811	ISR G1	2× FastEthernet, 4 HWIC slots	Voice/security module integration
1941	ISR G2	2× GigabitEthernet	Standard CCNA routing labs (OSPF/EIGRP)
2901	ISR G2	2× GigabitEthernet, 4 HWIC	Multi-branch enterprise simulation
2911/2921/2951	ISR G2	3× GigabitEthernet	Advanced routing, higher throughput labs
819	ISR (SOHO)	Integrated switch + WAN	Home/small office simulation
PT-Router (Generic)	Simplified	Configurable	Rapid prototyping without hardware constraints

Table 2.1: Router Model Comparison Matrix

2.1.3 Differentiating Factor: Fixed vs. Modular Routers

Table 2.2: Fixed-Configuration vs Modular Routers

Criteria	Fixed (e.g., 1841, 1941)	Modular (e.g., 2811, 2901+)
Port Expansion	Not possible	HWIC/NM/SM slots available
Cost/Complexity	Lower, simpler labs	Higher, supports voice/security modules
Use Case	CCNA fundamentals	CCNP-level or enterprise simulation

2.2 Switches

2.2.1 Functional Overview

Switches operate primarily at **Layer 2**, forwarding frames based on learned MAC addresses stored in the Content Addressable Memory (CAM) table. Multilayer switches additionally perform Layer 3 routing between VLANs.

2.2.2 Switch Model Specifications

Model	OSI Layer	Key Features	Use Case
2950-24	Layer 2	Basic VLAN, STP	Legacy CCNA labs
2960-24TT	Layer 2	VLAN, Trunking (802.1Q), STP, Port Security	Standard CCNA switching labs
3560-24PS	Layer 3	Inter-VLAN routing, static/dynamic routing	SVI configuration, routing+switching combined
3650-24PS	Layer 3	Advanced QoS, higher throughput	Enterprise campus simulation
Multilayer Switch (Generic)	Layer 3	Fully configurable routing/switching	Custom advanced topologies

Table 2.3: Switch Model Comparison Matrix

2.2.3 CLI Example: Basic VLAN Configuration on a 2960 Switch

Listing 2.1: VLAN and Trunk Configuration

```
1 Switch> enable
2 Switch# configure terminal
3 Switch(config)# vlan 10
4 Switch(config-vlan)# name SALES
5 Switch(config-vlan)# exit
```

```

6 Switch(config)# interface fastEthernet 0/1
7 Switch(config-if)# switchport mode access
8 Switch(config-if)# switchport access vlan 10
9 Switch(config-if)# exit
10 Switch(config)# interface fastEthernet 0/24
11 Switch(config-if)# switchport mode trunk

```

2.3 Hubs and Repeaters

Table 2.4: Hub vs Switch vs Repeater — Layer 1/2 Comparison

Attribute	Hub	Repeater	Switch
OSI Layer	1	1	2
Ports	Multiple (4–24)	2	Multiple (8–48)
Collision Domain	Single (shared)	Extended single	Per-port (separate)
Traffic Handling	Broadcasts to all ports	Regenerates signal	MAC-based forwarding
Duplex Mode	Half only	Half only	Full duplex supported

2.4 Wireless Infrastructure

2.4.1 Device Comparison

Device	OSI Layer	Function
Wireless Router (WRT300N)	L2/L3	Combines routing, NAT, DHCP server, switching, and wireless AP in a single unit — ideal for SOHO simulation.
Access Point (AP-PT)	L2	Pure wireless bridge; extends an existing wired LAN to wireless clients. Requires external DHCP/router.
AP-PT-N	L2	Higher-throughput AP supporting 802.11n standard clients.

Table 2.5: Wireless Device Comparison

NOTE

A common configuration mistake is treating an Access Point as a router. Access Points **do not** perform NAT or DHCP by default — these functions must be provided by an upstream router or server.

CHAPTER 3

End Devices

3.1 Overview

End devices represent the origination or termination point of network communication. They do not forward traffic on behalf of other devices (except when configured as a routing-capable server, which is atypical).

3.2 Device Specifications

Device	Connectivity	Config Tabs	Primary Use
PC	Wired (Ethernet)	Desktop, Config, Physical	IP configuration, ping/traceroute, terminal access
Laptop	Wired + Wireless	Desktop, Config, Physical	Mobility testing; wireless card (PCMCIA) simulation
Server	Wired	Services, Desktop, Config	Hosts DHCP, DNS, HTTP, TFTP, Email, FTP, AAA
IP Phone	Wired (PoE)	Config	VoIP simulation with Cisco CallManager Express
Smartphone	Wireless only	App, Desktop	Mobile app/browser simulation, WLAN testing
TabletPC	Wireless only	App, Desktop	Simplified mobile OS simulation
Printer	Wired/Wireless	Config	Shared network resource simulation

Table 3.1: End Device Specification Table

3.3 Server Services Reference

Table 3.2: Services Configurable on a Packet Tracer Server

Service	Description
HTTP/HTTPS	Hosts a basic website accessible via browser on client PCs
DHCP	Automatically assigns IP configuration to clients
DNS	Resolves domain names to IP addresses
TFTP	Stores IOS images/configuration files for router/switch backup
Email (POP3/SMTP)	Simulates email server-client communication
FTP	File transfer service between clients and server
AAA	Authentication, Authorization, Accounting for network access control

CHAPTER 4

IoT Components

4.1 Overview

Introduced in Packet Tracer 7.x, IoT components simulate smart-home and industrial automation scenarios, integrating programmable logic boards, sensors, and actuators controlled through an IoT server or Blockly/Python scripting.

4.2 Programmable Boards

Table 4.1: MCU-PT vs SBC-PT Comparison

Attribute	MCU-PT (Microcontroller)	SBC-PT (Single Board Computer)
Real-world Analog Processing Power	Arduino Uno Low (simple I/O logic)	Raspberry Pi Higher (can run scripts, OS-level tasks)
Programming Method	Blockly visual logic	Python / Blockly
Typical Use	Simple sensor-actuator triggers	Complex automation, data logging

4.3 Sensors (Input Devices)

Sensor	Function
Motion Detector (MD-PT)	Detects physical movement within range; commonly triggers lighting or alarm actuators
Fire/Smoke Detector	Detects smoke particles; triggers sprinkler/alarm systems
Water Level Sensor	Monitors fluid levels; triggers pump activation

Sensor	Function
Temperature/Humidity Sensor	Reports environmental data for HVAC automation
CO Detector	Detects carbon monoxide concentration for safety automation

Table 4.2: IoT Sensor Reference Table

4.4 Actuators (Output Devices)

Actuator	Function
Smart Light Bulb	Remote on/off/dimming control
Siren	Audible alarm output triggered by sensor conditions
Smart Plug	Remote power control for connected appliances
Garage Door	Automated open/close mechanism
Sprinkler/Pump	Water flow control triggered by fire/water sensors

Table 4.3: IoT Actuator Reference Table

Sensor–Actuator Logic Flow

Example Rule: *IF* Temperature Sensor > 30°C *THEN* activate Fan (Actuator). This condition-action logic is configured either directly on the IoT Server, or programmatically on an MCU/SBC board using Blockly or Python.

CHAPTER 5

Connections and Cabling Standards

5.1 Cable Type Reference

Cable Type	OSI Layer	Application
Straight-through (Copper)	1	Connects dissimilar devices (PC↔Switch, Switch↔Router)
Crossover (Copper)	1	Connects similar devices (PC↔PC, Switch↔Switch) without Auto-MDIX
Console (Rollover)	1	Out-of-band CLI access from PC to router/switch console port
Fiber Optic	1	High-speed, long-distance, EMI-immune connections
Serial (DTE/DCE)	1/2	Simulated WAN point-to-point links
Coaxial	1	Legacy cable modem connections
Phone (RJ-11)	1	Analog telephony / DSL modem connections

Table 5.1: Cable Types and Applications

5.2 Serial DCE/DTE Clocking Behavior

NOTE

In a WAN simulation using serial cables, the **DCE end** must have a clock rate manually configured; otherwise, the link will remain administratively down at Layer 1.

Listing 5.1: Configuring Clock Rate on the DCE Interface

```
1 Router(config)# interface serial 0/0/0
2 Router(config-if)# clock rate 64000
3 Router(config-if)# no shutdown
```

5.3 Straight-Through vs Crossover Decision Table

Table 5.2: Cable Selection Logic

Connection Scenario	Cable Required
PC → Switch	Straight-through
Switch → Router	Straight-through
PC → PC	Crossover
Switch → Switch	Crossover
Router → Router (Ethernet)	Crossover

CHAPTER 6

WAN Emulation and Cloud Devices

6.1 Purpose

WAN emulation devices simulate real-world Internet Service Provider (ISP) connectivity, allowing multiple independently addressed networks to interconnect without requiring a full physical WAN infrastructure model.

6.2 Device Reference

Device	Function
Cloud-PT	Central WAN simulation node; supports Ethernet, Serial, DSL, Cable, and Coaxial port modules for flexible ISP emulation
DSL Modem	Provides simulated broadband connectivity to the Cloud device via phone line
Cable Modem	Provides simulated broadband connectivity via coaxial cable

Table 6.1: WAN Emulation Devices



Figure 6.1: WAN Emulation Using Cloud-PT Between Two Independent Networks

CHAPTER 7

Multuser Connectivity

7.1 Overview

Packet Tracer’s **Multuser (MU) feature** enables two or more instances of the application — potentially on separate physical machines — to be linked together, allowing collaborative topology building across a real network connection (LAN/Internet).

7.2 Multuser vs Cloud Emulation

Table 7.1: Differentiating Multuser Connection from Cloud-PT WAN Emulation

Attribute	Multuser Connection	Cloud-PT
Scope	Connects separate .pkt files/instances	Simulates WAN within a single .pkt file
Use Case	Collaborative/instructor-student labs	Standalone WAN topology simulation
Requires Network Access	Yes (real IP connectivity between hosts)	No (fully virtual)

Typical Use Case

An instructor hosts Router A’s configuration on Machine 1; a student on Machine 2 configures Router B. Both machines run Packet Tracer simultaneously, connected via a Multuser Connection object, simulating a real inter-campus WAN link for joint troubleshooting exercises.

CHAPTER 8

Legacy Devices

8.1 Overview

Legacy devices are retained within Packet Tracer for historical/educational continuity, representing pre-modern networking equipment.

Device	OSI Layer	Description
Repeater	1	Two-port signal regenerator; extends maximum cable distance
Bridge	2	Software-based frame forwarding between two LAN segments; precursor to hardware switches
1900 Series Switch	2	Legacy Catalyst switch with limited VLAN/STP capability
800 Series Router	3	Legacy SOHO router used in older WAN topology labs

Table 8.1: Legacy Device Reference Table

8.2 Bridge vs Switch — Technical Distinction

Table 8.2: Bridge vs Modern Switch

Attribute	Bridge	Switch
Forwarding Method	Software-based (CPU)	Hardware-based (ASIC)
Port Density	2–4 ports	8–48+ ports
Speed	Store-and-forward only, slower	Wire-speed switching

CHAPTER 9

Networking Commands Reference — Basic to Advanced

9.1 Introduction

Effective network configuration and troubleshooting requires fluency in two categories of command-line environments:

1. **Cisco IOS CLI** — used to configure and verify routers, switches, and other Cisco infrastructure devices within Packet Tracer.
2. **Operating System Terminal** — used on end devices (PC/Laptop Desktop → Command Prompt, or a real OS terminal) to test connectivity, resolve names, and inspect host-level network configuration.

This chapter organizes commands progressively from *basic* to *advanced*, followed by a dedicated terminal command reference and a cross-platform comparison table.

9.2 Cisco IOS Command Modes

Mode	Prompt	Purpose
User EXEC	Router>	Limited monitoring commands only (e.g., <code>ping</code> , <code>show</code> basic status)
Privileged EXEC	Router#	Full monitoring access, including <code>show running-config</code>
Global Configuration	Router(config)#	System-wide settings (hostname, banner, routing)
Interface Configuration	Router(config-i)#	Settings applied to a specific interface
Line Configuration	Router(config-line)#	Console/VTY/AUX line settings (passwords, timeouts)
Router Configuration	Router(config-r)#	Dynamic routing protocol parameters

Table 9.1: Cisco IOS Hierarchical Command Modes

9.2.1 Navigating Between Modes

Listing 9.1: Mode Navigation Basics

```

1 Router> enable
2 Router# configure terminal
3 Router(config)# interface gigabitEthernet 0/0
4 Router(config-if)# exit
5 Router(config)# line vty 0 4
6 Router(config-line)# exit
7 Router(config)# exit
8 Router# disable

```

9.3 Basic-Level Commands (Foundational)

Command	Description
enable	Enters Privileged EXEC mode
configure terminal	Enters Global Configuration mode
hostname <name>	Sets the device's hostname
interface <type><number>	Enters configuration mode for a specific interface
ip address <ip> <mask>	Assigns an IPv4 address to an interface
no shutdown	Administratively enables an interface
exit / end	Moves up one config level / returns to Privileged EXEC
write memory or copy running-config startup-config	Saves current configuration to NVRAM
erase startup-config	Deletes saved configuration (factory reset on reload)

Table 9.2: Basic Cisco IOS Commands

Listing 9.2: Minimal Router Bring-Up Sequence

```

1 Router> enable
2 Router# configure terminal
3 Router(config)# hostname R1
4 R1(config)# interface gigabitEthernet 0/0
5 R1(config-if)# ip address 192.168.1.1 255.255.255.0
6 R1(config-if)# no shutdown
7 R1(config-if)# end
8 R1# copy running-config startup-config

```

9.4 Intermediate-Level Commands

Command	Description
<code>vlan <id></code>	Creates/edits a VLAN on a switch
<code>switchport mode access/trunk</code>	Sets a port as access or trunk
<code>switchport access vlan <id></code>	Assigns an access port to a VLAN
<code>ip dhcp pool <name></code>	Creates a DHCP address pool on a router
<code>network <net> <wildcard></code>	Advertises a network under a routing protocol
<code>banner motd #message#</code>	Sets a login banner message
<code>enable secret <password></code>	Sets an encrypted privileged mode password
<code>service password-encryption</code>	Encrypts all plaintext passwords in config
<code>line vty 0 4 / password / login</code>	Configures Telnet/SSH remote access

Table 9.3: Intermediate Cisco IOS Commands

Listing 9.3: DHCP Pool and Remote Access Configuration

```

1 Router(config)# ip dhcp pool LAN_POOL
2 Router(dhcp-config)# network 192.168.1.0 255.255.255.0
3 Router(dhcp-config)# default-router 192.168.1.1
4 Router(dhcp-config)# dns-server 8.8.8.8
5 Router(dhcp-config)# exit
6 Router(config)# line vty 0 4
7 Router(config-line)# password cisco123
8 Router(config-line)# login
9 Router(config-line)# transport input ssh

```

9.5 Advanced-Level Commands

Command	Description
<code>router ospf <process-id></code>	Enables OSPF routing process
<code>router eigrp <as-number></code>	Enables EIGRP routing process
<code>access-list <num> permit/deny ...</code>	Defines a standard/extended ACL rule

Command	Description
<code>ip nat inside/outside</code>	Marks interfaces for NAT translation direction
<code>ip nat inside source list <acl> interface <if> overload</code>	Configures PAT (NAT overload)
<code>switchport port-security</code>	Enables port security on an interface
<code>switchport port-security maximum <n></code>	Limits number of allowed MAC addresses
<code>standby <group> ip <vip></code>	Configures HSRP virtual gateway
<code>crypto key generate rsa</code>	Generates RSA keys required for SSH

Table 9.4: Advanced Cisco IOS Commands

Listing 9.4: OSPF Routing and NAT Overload Configuration

```

1 Router(config)# router ospf 1
2 Router(config-router)# network 192.168.1.0 0.0.0.255 area 0
3 Router(config-router)# network 10.0.0.0 0.0.0.3 area 0
4 Router(config-router)# exit
5
6 Router(config)# access-list 1 permit 192.168.1.0 0.0.0.255
7 Router(config)# interface gigabitEthernet 0/0
8 Router(config-if)# ip nat inside
9 Router(config-if)# exit
10 Router(config)# interface serial 0/0/0
11 Router(config-if)# ip nat outside
12 Router(config-if)# exit
13 Router(config)# ip nat inside source list 1 interface serial 0/0/0
    overload

```

Listing 9.5: Enabling SSH Access on a Router

```

1 Router(config)# hostname R1
2 R1(config)# ip domain-name lab.local
3 R1(config)# username admin secret StrOngPass!
4 R1(config)# crypto key generate rsa
5 % Choose modulus size: 1024
6 R1(config)# line vty 0 4
7 R1(config-line)# transport input ssh
8 R1(config-line)# login local

```

9.6 Verification and Troubleshooting Commands

Command	Description
<code>show running-config</code>	Displays the active configuration in RAM
<code>show startup-config</code>	Displays the saved configuration in NVRAM
<code>show ip interface brief</code>	Summarizes interface status and IP addresses
<code>show ip route</code>	Displays the current IP routing table
<code>show vlan brief</code>	Lists VLANs and their assigned ports
<code>show interfaces <if></code>	Detailed interface statistics (errors, drops, duplex)
<code>show cdp neighbors</code>	Displays directly connected Cisco devices
<code>show mac address-table</code>	Displays the switch's learned MAC-to-port mapping
<code>debug ip routing</code>	Real-time logging of routing table changes
<code>ping <ip></code>	Tests Layer 3 reachability from the router
<code>traceroute <ip></code>	Displays the hop-by-hop path to a destination

Table 9.5: Cisco IOS Verification and Troubleshooting Commands

NOTE

Always pair a configuration change with a verification command. For example, after configuring OSPF, run `show ip ospf neighbor` to confirm adjacency formation before assuming the protocol is functioning correctly.

9.7 Operating System Terminal Commands

End devices (PC/Laptop) in Packet Tracer provide a simplified *Command Prompt* under the Desktop tab. The commands below mirror real-world Windows, Linux, and macOS terminal usage for network diagnostics.

9.7.1 Windows Command Prompt

Listing 9.6: Windows Networking Commands

```
C:\> ipconfig /all
C:\> ping 192.168.1.1
C:\> tracert google.com
C:\> nslookup google.com
C:\> netstat -an
C:\> arp -a
C:\> route print
C:\> telnet 192.168.1.1
C:\> ssh admin@192.168.1.1
```

9.7.2 Linux / macOS Terminal

Listing 9.7: Linux/macOS Networking Commands

```
$ ifconfig
$ ip addr show
$ ping -c 4 192.168.1.1
$ traceroute google.com
$ dig google.com
$ netstat -tulnp
$ arp -a
$ ip route show
$ ssh admin@192.168.1.1
$ curl -I http://example.com
```

9.7.3 Packet Tracer PC Command Prompt (Simplified Subset)

Listing 9.8: Commands Available on Packet Tracer End-Device Terminal

```
PC> ipconfig
PC> ipconfig /all
PC> ping 192.168.1.1
PC> tracert 8.8.8.8
PC> arp -a
PC> nslookup cisco.com
PC> telnet 192.168.1.1
PC> ssh -l admin 192.168.1.1
```

NOTE

Packet Tracer’s simulated terminal supports only a subset of real OS commands. Options such as `-c` (Linux ping count) or advanced `netstat` flags are **not** supported and will return a syntax error.

9.8 Cross-Platform Command Comparison

Purpose	Cisco IOS	Windows	Linux/macOS
View IP configuration	<code>show ip interface brief</code>	<code>ipconfig /all</code>	<code>ifconfig / ip addr</code>
Test reachability	<code>ping <ip></code>	<code>ping <ip></code>	<code>ping -c 4 <ip></code>
Trace network path	<code>tracert <ip></code>	<code>tracert <ip></code>	<code>tracert <ip></code>
Resolve domain name	<code>ip domain-lookup</code>	<code>nslookup <name></code>	<code>dig <name> / nslookup</code>
View routing table	<code>show ip route</code>	<code>route print</code>	<code>ip route show</code>
View ARP cache	<code>show arp</code>	<code>arp -a</code>	<code>arp -a</code>
Remote CLI access	<code>line vty (SSH/Telnet)</code>	<code>ssh / telnet</code>	<code>ssh / telnet</code>
View active connections	<code>show tcp brief</code>	<code>netstat -an</code>	<code>netstat -tulnp</code>

Table 9.6: Command Equivalence Across Cisco IOS, Windows, and Linux/macOS

Practical Tip

When troubleshooting end-to-end connectivity in Packet Tracer, follow this diagnostic order:
1. `ipconfig` on PC (verify local IP/gateway) → **2.** `ping` default gateway → **3.** `show ip route` on router (verify routing) → **4.** `ping` remote host → **5.** `tracert`/`tracert` to isolate the failing hop.

CHAPTER 10

Consolidated Decision Matrix

10.1 Device Selection Guide by Networking Requirement

Requirement	Recommended Device	Justification
Connect multiple hosts, single broadcast domain	Switch (2960)	Layer 2 MAC-based forwarding, full duplex
Route between subnets	Router (1941/2911)	Layer 3 routing table processing
Inter-VLAN routing without separate router	Multilayer Switch (3560/3650)	SVI-based Layer 3 switching
Provide DHCP/DNS/Web services	Server	Dedicated services engine
Home network with WiFi	Wireless Router	Integrated NAT + DHCP + AP
Extend wired LAN wirelessly	Access Point	Layer 2 bridge, no routing overhead
CLI configuration access	Console Cable + PC	Out-of-band management
Simulate long-distance WAN	Serial Cable (DTE/DCE) + Cloud-PT	Point-to-point WAN emulation
Build home automation lab	MCU/SBC + Sensors + Actuators	Programmable I/O logic
Cross-machine collaborative lab	Multiuser Connection	Bridges independent PT sessions

Table 10.1: Master Decision Matrix for Device Selection

APPENDIX A

CLI Command Quick Reference Cards

Listing A.1: Basic Router Interface and Routing Configuration

```
1 Router> enable
2 Router# configure terminal
3 Router(config)# hostname R1
4 R1(config)# interface gigabitEthernet 0/0
5 R1(config-if)# ip address 192.168.1.1 255.255.255.0
6 R1(config-if)# no shutdown
7 R1(config-if)# exit
8 R1(config)# router ospf 1
9 R1(config-router)# network 192.168.1.0 0.0.0.255 area 0
```

Listing A.2: Switch Port Security Configuration

```
1 Switch(config)# interface fastEthernet 0/1
2 Switch(config-if)# switchport mode access
3 Switch(config-if)# switchport port-security
4 Switch(config-if)# switchport port-security maximum 2
5 Switch(config-if)# switchport port-security violation shutdown
```

Listing A.3: Quick Terminal Diagnostic Sequence (PC)

```
PC> ipconfig /all
PC> ping 192.168.1.1
PC> tracert 8.8.8.8
PC> nslookup cisco.com
```

APPENDIX B

Glossary of Terms

Term	Definition
ACL	Access Control List — filters traffic based on defined rules
CAM Table	Content Addressable Memory table storing learned MAC addresses on a switch
DCE/DTE	Data Communication Equipment / Data Terminal Equipment — defines clocking role in serial WAN links
NAT	Network Address Translation — maps private IPs to public IPs
SVI	Switch Virtual Interface — virtual Layer 3 interface on a multilayer switch
VLAN	Virtual Local Area Network — logical segmentation of a broadcast domain
CLI	Command Line Interface — text-based interface for device configuration
PAT	Port Address Translation — NAT overload mapping multiple private IPs to one public IP using ports
HSRP	Hot Standby Router Protocol — provides gateway redundancy

Table B.1: Glossary of Networking Terms